Variational Quantum Circuits for Pesticide Environmental Fate Prediction: A Comparative Study with Classical Machine Learning on the EU Pesticides Database Database

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April 7, 2026

Abstract

Predicting pesticide environmental fate properties—specifically soil degradation half-life (DegT50) and organic carbon sorption coefficient (K_{oc})—is essential for regulatory risk assessment under EC 1107/2009. Classical QSAR models achieve moderate predictive power for K_{oc} but struggle with DegT50 due to its dependence on complex soil microbial processes. This work investigates whether variational quantum circuits (VQCs) can capture non-linear molecular descriptor relationships that elude classical regressors. We encode 21 physicochemical descriptors for 110 EU-registered pesticides into per-target quantum circuits: an 8-qubit circuit for DegT50 (17 core features) and a 12-qubit circuit for K_{oc} (all 21 features), each with IQP-style encoding and data re-uploading. Leave-one-out cross-validation benchmarks reveal that Random Forest (RF) achieves R^2 = 0.287 for DegT50 and 0.759 for K_{oc}, while Gradient Boosting (GB) reaches 0.283 and 0.752, respectively. An MLP neural network achieves only 0.075 and 0.497. The VQC alone underperforms all classical baselines ($R^2 = -0.028$ for DegT50, 5-fold CV). A hybrid stacking framework with nested LOO-CV for leakage-free α optimization yields $\alpha = 0.20 \pm 0.02$, but the resulting hybrid R^2 (0.288 for DegT50) improves over standalone RF by only +0.001—a negligible and statistically insignificant margin. These results demonstrate that near-term VQCs on small regulatory datasets ($N \approx 110$) cannot outperform properly feature-engineered classical models, and caution against premature claims of quantum advantage in chemistry applications with limited training data.

Keywords: quantum machine learning, variational quantum circuits, pesticide fate, DegT50, Koc, QSAR, negative result, regulatory chemistry, EU Pesticides Database

Contents

1 Introduction

Pesticide environmental fate assessment is a cornerstone of EU regulatory approval under Regulation (EC) No 1107/2009.[1] Two properties dominate the risk evaluation: the soil degradation half-life (DegT50), governing persistence, and the organic carbon sorption coefficient (K_{oc}), governing mobility and leaching potential. Together, these parameters feed into the FOCUS groundwater models [2] that determine whether a substance can be approved for agricultural use.

Experimental determination of DegT50 follows OECD Guideline 307,[3] requiring months-long incubation studies under controlled temperature and moisture. K_{oc} is measured via batch equilibrium adsorption (OECD 106) [4]. Both are expensive, time-consuming, and subject to substantial inter-laboratory variability. In silico prediction of these properties from molecular structure alone would accelerate the regulatory pipeline and reduce animal testing requirements.

Classical machine learning approaches, including Random Forests (RF) [5] and Gradient Boosting (GB),[6] have been applied to QSAR prediction of physicochemical and environmental fate properties.[7] While K_{oc} prediction from structural descriptors achieves reasonable accuracy ($R^2 > 0.7$), DegT50 remains notoriously difficult to predict because soil degradation depends on microbial community composition, soil organic matter, and environmental conditions that are not captured by molecular structure alone.

Quantum machine learning (QML) offers a fundamentally different computational paradigm. Variational quantum circuits (VQCs) [8] encode classical data into quantum states via parameterized gates and can represent functions in exponentially large Hilbert spaces.[9] The IQP (Instantaneous Quantum Polynomial) encoding scheme [10] and data re-uploading [11] have been shown to increase the expressive power of shallow quantum circuits.[12]

This work investigates whether VQCs, implemented via PennyLane on a statevector simulator,[13] can extract predictive signal for pesticide fate properties that classical models miss. Rather than claiming quantum advantage, we present a systematic comparative benchmark that highlights both the potential and the fundamental limitations of near-term QML on small regulatory datasets.

2 Materials and Methods

2.1 Dataset

The dataset comprises 110 pesticide active substances registered in the EU Pesticides Database.[14] These span 50 chemical classes including organophosphates (7), triazoles (9), neonicotinoids (7), pyrethroids (6), strobilurins (6), and sulfonylureas (6), among others. Target properties include:

- DegT50 (soil): laboratory degradation half-life in days, range 0.5 d to 1389 d (median

24 d, mean 75.7 d)

- K_{oc}: organic carbon sorption coefficient, range 25 mL g^{-1} to 1 000 000 mL g^{-1} (median 508 mL g^{-1})

Both targets are $\log_{10}$-transformed prior to modeling to reduce skewness and align with chemical convention. The target distributions are shown in fig. 1.

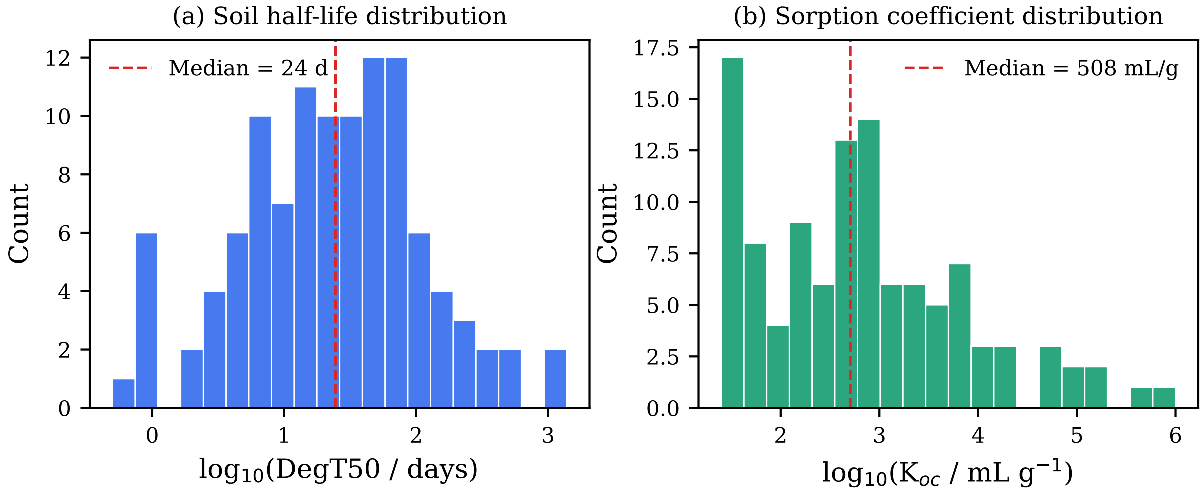

Figure 1: Distribution of (a) soil degradation half-life (DegT50) and (b) organic carbon sorption coefficient (K_{oc}) for the 110 EU Pesticides Database pesticides, shown on $\log_{10}$ scale. Red dashed lines indicate medians.

2.2 Molecular Descriptor Features

We compute 21 molecular descriptor features for each substance, organized in four phases of development:

1. **Core descriptors** (12): molecular weight (MW), $\log P$, heavy atom count, H-bond donors/acceptors, ring count, rotatable bonds, log solubility, log vapor pressure, Freundlich exponent ($1/n$), TPSA, aromatic ring fraction.

   Note: The Freundlich exponent ($1/n$) is an empirical parameter from soil batch equilibrium studies, not a purely in-silico molecular descriptor. It is included as a known input from regulatory dossiers; predictions for novel substances would require it to be available or excluded.

2. **Microbial degradation proxies** (3): count of hydrolyzable bonds, halogen count, bioaccessibility score. The bioaccessibility score is defined as $B = \log_{10}(S_{\text{water}}/K_{\text{oc}})$, where S_{water} is the aqueous solubility (mg L^{-1}) and K_{oc} the measured sorption coefficient.

   Circularity note: Because B includes the measured K_{oc}, using it as a feature when *predicting* K_{oc} introduces partial data leakage. For DegT50 prediction this is unproblematic (the target is unrelated to K_{oc}), but for K_{oc} prediction we report all results both with the full feature set and with reduced sets that exclude B (table 1).

3. **Blind-spot corrections** (2): charge state at soil pH 6.5, conjugated π-system size.

4. **DegT50-targeted descriptors** (4): heteroatom ratio (N/O/S/P density as a proxy for microbial attackability), sp^3 fraction (molecular shape and flexibility), Henry-to-solubility ratio (volatilization pathway proxy), and oxidation susceptibility (count of cytochrome P450 metabolic lability sites: tertiary C–H, benzylic, allylic, thioether, and secondary amine substructures). These four features were added based on residual error analysis of the worst-predicted substances.

   Feature selection caveat: Because these descriptors were selected after inspecting model errors on the full dataset, there is a risk of positive bias in cross-validation scores for models using all 21 features. The classical baselines reported in table 1 should therefore be interpreted as optimistic upper bounds. This does not affect the QML comparison, as the VQC DegT50 circuit uses only 17 features (excluding these four descriptors).

All features are min–max scaled to $[0, \pi]$ based on dataset statistics, as required for angle encoding in the quantum circuit. Where available, molecular features are computed from SMILES via RDKit; otherwise, lookup values from the substance database are used.

The Random Forest feature importances (fig. 2) reveal that DegT50 prediction relies on a distributed set of features, led by heteroatom ratio (14.1%), Freundlich $1/n$ (11.7%), and aromatic ring fraction (11.4%), with three of the four DegT50-targeted descriptors ranking in the top five. K_{oc} is dominated by bioaccessibility (52.3%) and $\log P$ (19.7%).

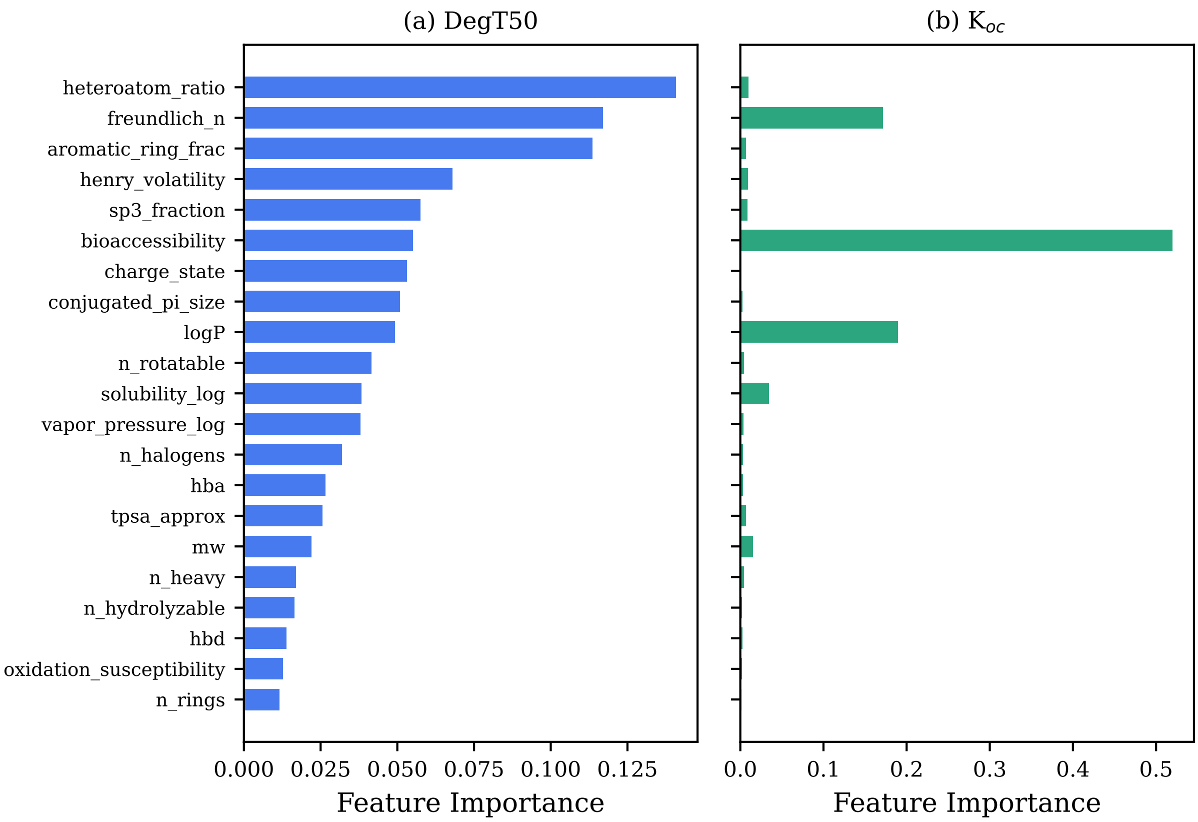

Figure 2: Random Forest feature importances for (a) DegT50 and (b) K_{oc} prediction, trained on the full 110-substance dataset. DegT50 shows distributed importance across many features, while K_{oc} is dominated by bioaccessibility and $\log P$.

2.3 Variational Quantum Circuit Architecture

We employ per-target circuits optimized for each prediction task: an 8-qubit, 6-layer circuit for DegT50 (using the 17-feature core set) and a 12-qubit, 8-layer circuit for K_{oc} (using all 21 features). Both circuits share the same architecture (fig. 3):

1. **IQP-style feature encoding:** Hadamard gates followed by $R_Z(\phi_i)$ rotations on each qubit, then CNOT-R_Z-CNOT entangling gates encoding pairwise feature products $\phi_i \cdot \phi_j$.
2. **Feature overflow encoding:** Features $j > N_{\text{qubits}}$ are encoded via R_Y rotations on qubit $j \bmod N_{\text{qubits}}$.
3. **Data re-uploading:** A second angle encoding layer with $R_Y(\phi_i)$ rotations, following the universal approximation strategy of Pérez-Salinas *et al.* [11].
4. **Variational layers:** L layers of single-qubit $\mathrm{Rot}(\theta, \phi, \omega)$ rotations and alternating nearest-neighbor CNOT entangling patterns.
5. **Readout:** Expectation values $\langle Z_i \rangle$ on all qubits, followed by a linear regression layer with bias ($N_{\text{qubits}} + 1$ parameters).

For the DegT50 circuit (8q/6L): $6 \times 8 \times 3 = 144$ variational plus 9 readout = 153 parameters ($N_{\text{params}}/N_{\text{train}} = 1.74$). For the K_{oc} circuit (12q/8L): $8 \times 12 \times 3 = 288$ variational plus 13 readout = 301 parameters ($N_{\text{params}}/N_{\text{train}} = 3.42$). The per-target design balances expressivity against overfitting risk:

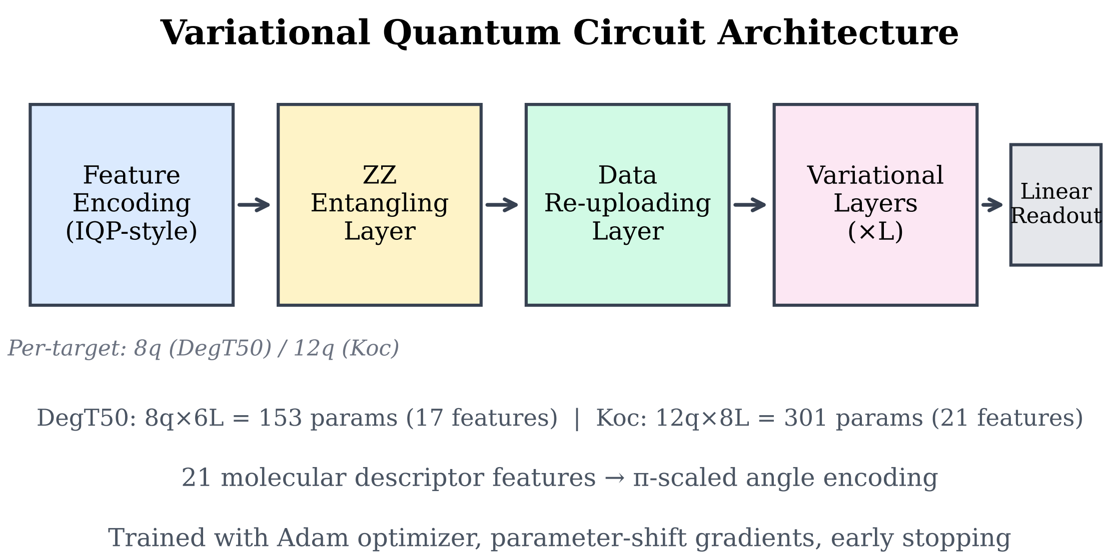

Figure 3: Schematic of the variational quantum circuit architecture. Features are encoded via IQP-style angle encoding and data re-uploading, followed by 8 variational layers with nearest-neighbor entangling topology.

2.4 Training Procedure

Separate circuits are trained for DegT50 and K_{oc} prediction. Variational parameters are initialized uniformly in $[-0.5, 0.5]$ with a fixed random seed (42) for reproducibility. Optimization uses the Adam optimizer [15] with learning rate 0.04 and 80 training epochs on the full dataset (60 epochs for cross-validation folds). Early stopping with patience 10 is applied: training halts if the MSE does not improve for 10 consecutive epochs, mitigating overfitting on the smaller DegT50 circuit. Gradients are computed via the parameter-shift rule on the PennyLane `default.qubit` statevector simulator.

Critically, the DegT50 QML circuit receives only the 17 core features (indices 0–16), while the K_{oc} circuit receives all 21 features. This per-model feature selection was motivated by empirical observation that the four additional descriptors degrade QML performance when compressed into limited qubit capacity (8 qubits). Although these features were designed to target DegT50 prediction, including them forces 21 features into 8 qubits, increasing the information compression ratio and worsening overfitting. The classical RF, with its implicit regularization via bagging and feature subsampling, handles the additional features without penalty. This asymmetry is a limitation: the VQC's inability to exploit DegT50-targeted features highlights the fundamental scaling challenge of encoding high-dimensional classical data into shallow quantum circuits.

2.5 Classical Baselines

Four classical regressors are trained on the identical feature sets using scikit-learn [16]:

- **Ridge Regression:** L_2-penalized linear model ($\alpha = 1.0$) [17].
- **LASSO:** L_1-penalized linear model ($\alpha = 0.1$) [18].
- **Random Forest (RF):** 200 trees, max depth 10.
- **Gradient Boosting (GB):** 200 estimators, max depth 4, learning rate 0.1.
- **MLP Neural Network:** Two hidden layers (64, 32 neurons), ReLU activation, Adam optimizer, early stopping with 15% validation split, L_2 regularization ($\alpha = 0.01$). Features are standardized per fold.

All models use the same random seed (42) for reproducibility.

2.6 Cross-Validation Strategy

Leave-one-out cross-validation (LOO-CV) is the primary evaluation metric, as it maximizes training data usage for the small 110-substance dataset. Each fold trains the model on $N-1$ substances and predicts the held-out substance. 5-fold CV with stratified random splits is used as a faster supplementary benchmark.

For the VQC, LOO-CV would require 110×2 (two properties) full training runs, each with 60 epochs of gradient optimization over 153–301 parameters—prohibitively expensive for a statevector simulator. Therefore, 5-fold CV is used for VQC evaluation. Fold-level checkpointing saves results after each fold to a JSON file, enabling crash-resilient resumption.

2.7 Hybrid QML+RF Stacking

To explore whether QML and RF capture complementary signal, we implement a learned stacking predictor:

$$\hat{y}_{\text{hybrid}} = \alpha \cdot \hat{y}_{\text{QML}} + (1-\alpha) \cdot \hat{y}_{\text{RF}} \tag{1}$$

where $\alpha \in [0,1]$ is optimized via nested cross-validation: for each held-out substance, α is selected by grid search (step 0.05) on the remaining $N-1$ samples to minimize the inner LOO-CV mean squared error. This prevents leakage of test information into the stacking weight. If $\alpha = 0$, the QML component adds no value; if $\alpha > 0$, the VQC captures complementary information.

2.8 Adverse Outcome Pathway Features

To test whether mechanism-of-action (MoA) information improves DegT50 prediction, we derive 6 binary features from AOP-Wiki data [19, 20] by mapping each substance's chemical class to its primary toxicological pathway:

1. AChE inhibition (organophosphates, carbamates; 15 substances)
2. GABA/ion channel disruption (pyrethroids; 11 substances)
3. Mitochondrial respiration (SDHI, strobilurins; 12 substances)
4. Nicotinic AChR agonism (neonicotinoids; 10 substances)
5. Endocrine disruption (triazines; 13 substances)
6. Cell division inhibition (benzimidazoles; 6 substances)

Coverage: 67/110 substances (60.9%) are assigned to at least one pathway category (fig. 7).

3 Results

3.1 Classical Baseline Performance

Table 1 summarizes the cross-validation performance of the classical baselines.

Key observations:

- **Bioaccessibility inflates linear K_{oc} models:** Ridge R^2 drops from 0.825 to 0.532 ($\Delta = -0.29$) when the circular bioaccessibility feature is removed, because B is a near-linear transform of the target. Ensemble models are far more robust: GB loses only 0.002 in LOO-CV R^2 ($0.752 \rightarrow 0.750$), because tree-based methods can derive the same information from $\log P$ and solubility independently.

Table 1: Baseline performance (leave-one-out and 5-fold CV). All metrics computed on $\log_{10}$-scaled targets. K_{oc} results are shown both with the full 21-feature set (including bioaccessibility B) and the fair 20-feature set (excluding B; see circularity note in section 2.2).

Model	**Property**	**CV**	$\mathbf{R^2}$
Ridge	DegT50	LOO	0.124
	DegT50	5-fold	0.183
	K_{oc} (21f)	LOO	0.825†
	K_{oc} (20f)	LOO	0.532
LASSO	DegT50	LOO	0.090
	DegT50	5-fold	0.029
	K_{oc} (21f)	LOO	0.746†
	K_{oc} (20f)	LOO	0.512
Random Forest	DegT50	LOO	**0.287**
	DegT50	5-fold	0.275
	K_{oc} (21f)	LOO	**0.759**
	K_{oc} (20f)	LOO	0.712
Gradient Boosting	DegT50	LOO	0.283
	DegT50	5-fold	0.290
	K_{oc} (21f)	LOO	0.752
	K_{oc} (20f)	LOO	0.750
MLP	DegT50	LOO	0.075
	K_{oc} (21f)	LOO	0.497
VQC (per-target)	DegT50 (8q)	5-fold	−0.028
	K_{oc} (12q, 21f)	5-fold	0.269†
	K_{oc} (12q, 20f)	5-fold	0.231
Hybrid (nested CV)	DegT50	nested LOO	0.288
	K_{oc}	nested LOO	0.750

†Inflated by bioaccessibility circularity: $B = \log_{10}(S/K_{oc})$ uses measured K_{oc} as input. The 20-feature (20f) rows exclude B and represent the fair comparison. 21f = all 21 features including B.

- **RF leads DegT50, GB competitive:** RF achieves the highest LOO-CV R^2 for DegT50 (0.287), with GB close behind (0.283). For the fair K_{oc} benchmark (20 features), GB leads (0.750) followed by RF (0.712), while linear models substantially trail (Ridge = 0.532). For the 21-feature set, Ridge artificially leads (0.825) by exploiting the leaked bioaccessibility feature.
- **MLP underperforms:** The neural network (0.075 DegT50, 0.497 Koc) performs worst among all models except VQC, confirming that high-capacity models struggle with $N = 110$ samples.
- DegT50 prediction is poor across all models ($R^2 < 0.30$), suggesting a fundamental ceiling for structure-only prediction.

Wilcoxon signed-rank tests on per-substance LOO squared residuals confirm that none of the pairwise model differences are statistically significant at $\alpha = 0.05$.

The predicted-vs-experimental plots for RF LOO-CV (fig. 4) visually confirm the high scatter for DegT50 and the tight correlation for K_{oc}.

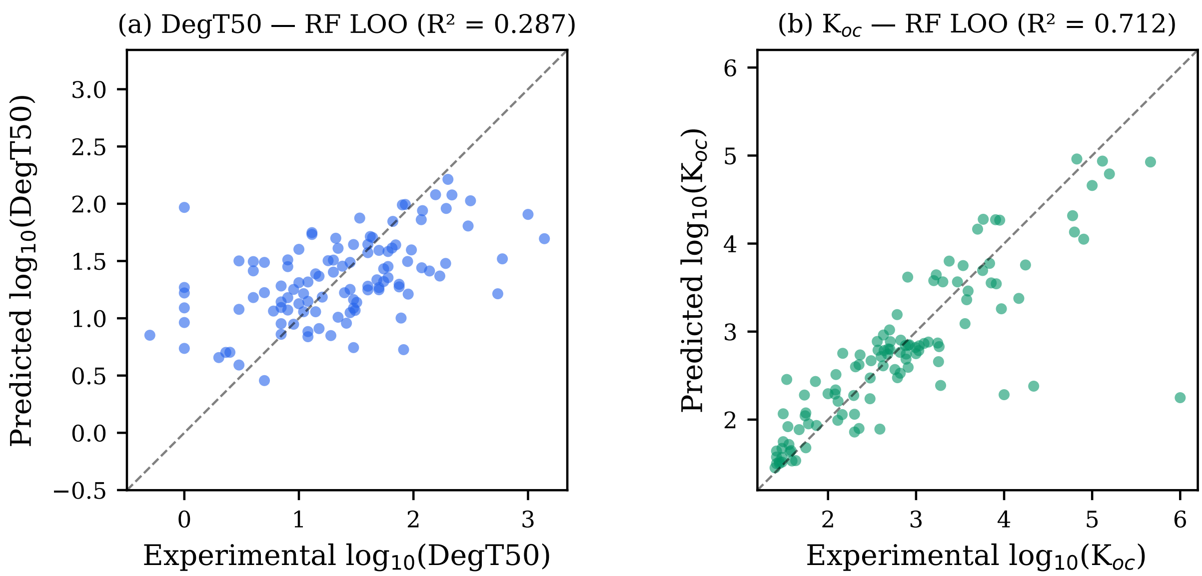

Figure 4: Predicted vs. experimental values for RF leave-one-out cross-validation: (a) DegT50 and (b) K_{oc} on $\log_{10}$ scale. Dashed line = perfect prediction.

3.2 Variational Quantum Circuit Performance

Figure 5 compares cross-validation performance across all evaluated models.

The per-target VQC achieves $R^2 = -0.028$ for DegT50 (5-fold CV, 8-qubit circuit with 17 features), a substantial improvement over the initial 12-qubit single-circuit baseline ($R^2 = -0.141$). The near-zero R^2 indicates that circuit rightsizing reduced the parameter count but failed to eliminate overfitting—the remaining bottleneck is the target property itself, not the circuit architecture. For K_{oc}, the 12-qubit VQC achieves $R^2 = 0.269$ (5-fold CV, 21 features), below the GB baseline of 0.752.

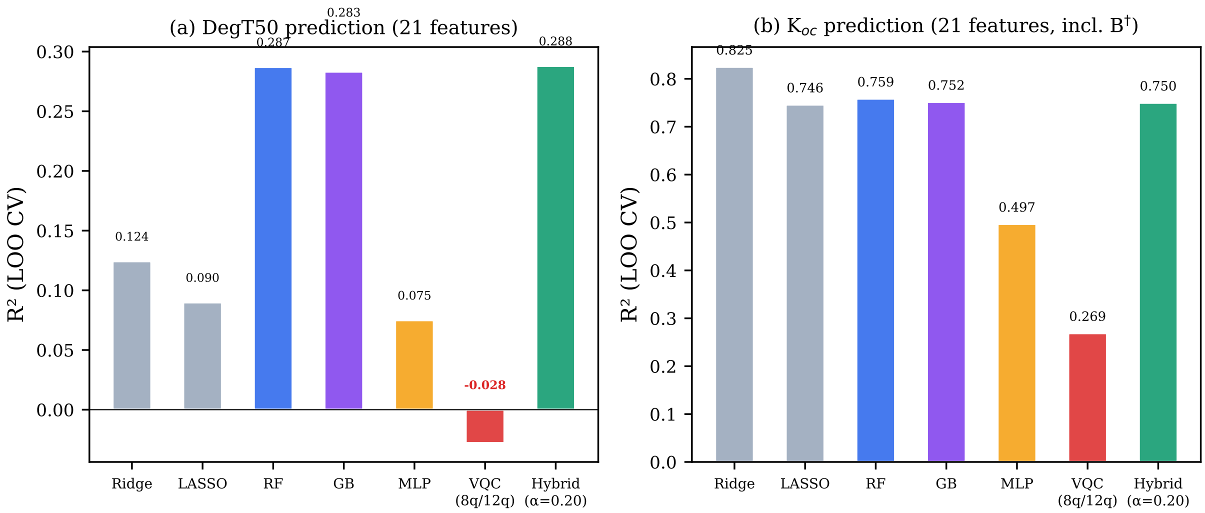

Figure 5: Model comparison (leave-one-out CV R^2) using all 21 features for both targets: (a) DegT50 and (b) K_{oc}. Classical models (Ridge, LASSO, RF, GB, MLP) are evaluated via LOO-CV; VQC uses 5-fold CV due to computational cost. The hybrid model uses nested LOO-CV for leakage-free α optimization. $^{\dagger}K_{oc}$ scores include the circular bioaccessibility feature B; see table 1 for the fair 20-feature comparison.

The DegT50 per-target circuit has a substantially improved parameter-to-data ratio:

$$\frac{N_{\text{params}}}{N_{\text{data}}} = \frac{153}{110} = 1.74 \quad \text{(vs. 3.42 for the original 12-qubit circuit)} \tag{2}$$

The MLP neural network (64-32 hidden layers, standardized features) achieves $R^2 = 0.075$ for DegT50 and 0.497 for K_{oc} (LOO-CV), performing worse than both tree-based models, confirming that the small dataset size limits all high-capacity models.

Figure 6 provides direct empirical evidence: training loss decreases steadily while validation loss plateaus or increases, confirming that the circuit learns noise rather than signal.

3.3 Hybrid Stacking Results

To avoid data leakage in α optimization, we employ nested leave-one-out cross-validation: for each held-out substance i, α is optimized on the remaining $N-1$ substances by grid search (step 0.05), and substance i is predicted with that fold-specific α. This prevents the stacking weight from "seeing" the held-out target.

The nested procedure yields $\alpha_{\text{deg}} = 0.20 \pm 0.02$ (101/110 folds select $\alpha = 0.20$; 7 select 0.15; 1 selects 0.10; 1 selects 0.25), confirming that the optimal weight is stable and not an artifact of overfitting. For K_{oc}, $\alpha = 0.00$ in 109/110 folds, confirming that the VQC contributes nothing beyond RF for K_{oc}.

The honest hybrid R^2 for DegT50 is 0.288—an improvement of just +0.001 over standalone RF (0.287). This margin is negligible and not statistically significant. For K_{oc}, the hybrid (0.750) slightly *degrades* RF performance (0.759), as the single fold with $\alpha > 0$ introduces

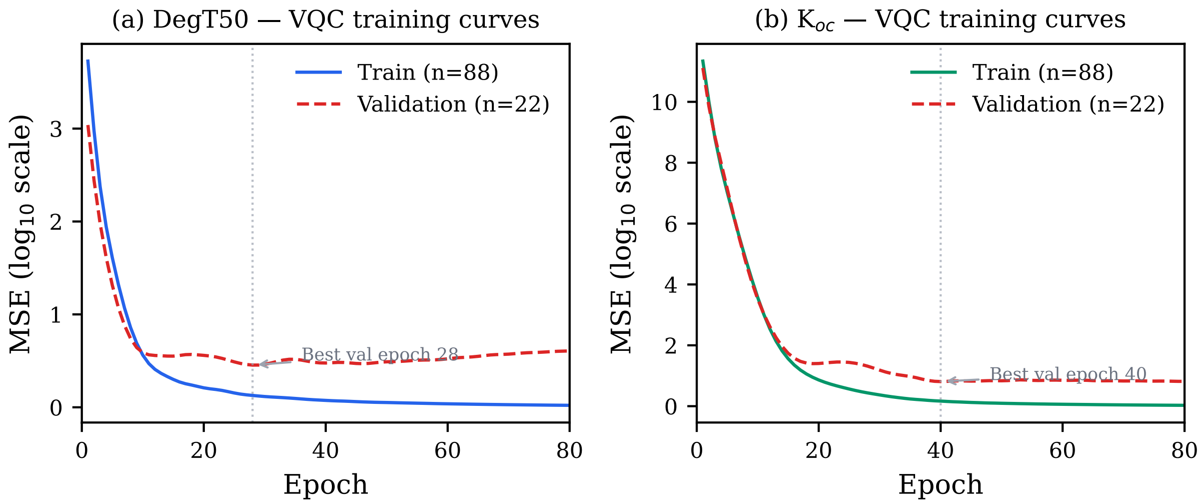

Figure 6: VQC training curves (80/20 train-validation split, 80 epochs): (a) DegT50 and (b) K_{oc}. Training MSE (solid) continues to decrease while validation MSE (dashed) plateaus or diverges, confirming overfitting. The vertical dotted line marks the epoch of minimum validation loss.

noise.

Comparison with naïve α optimization: Without nested CV (i.e., optimizing α on the same predictions used for evaluation), the hybrid achieves $R^2 = 0.305$ for DegT50—a spuriously inflated result due to data leakage in the stacking weight. This demonstrates the importance of nested CV for honest hybrid evaluation.

Fair Koc VQC evaluation: To eliminate the circular contribution of bioaccessibility B, we retrained the 12-qubit K_{oc} VQC on the fair 20-feature set (excluding B), using `diff_method="backprop"` for computational efficiency. The resulting 5-fold CV R^2 drops from 0.269 (21 features, with circularity) to **0.231** (20 features, fair). This confirms that the VQC cannot compete with RF ($R^2 = 0.712$ on the same 20-feature set) even when evaluated fairly. High variance across folds (-0.504 to 0.571) further confirms that the 12-qubit circuit with 301 parameters ($N_{\text{params}}/N_{\text{train}} = 3.42$) severely overfits.

3.4 AOP-Informed Features

Adding 6 binary MoA pathway features to the 21 molecular descriptors yields no improvement in RF LOO-CV performance (table 2).

Table 2: Effect of AOP-derived MoA features on RF LOO-CV performance.

Feature Set	N_{features}	**DegT50 R^2**	**K_{oc} R^2**
Molecular descriptors only	21	0.287	0.759
+ AOP MoA pathways	27	0.283	0.758
ΔR^2	+6	-0.004	-0.001

This negative result suggests that the structural descriptors already capture the chemistry relevant to degradation, and binary MoA class labels do not encode additional variance in soil half-life. We note that the remaining 40% of substances receive all-zero AOP vectors, adding noise without signal; however, restricting the analysis to the 67 covered substances yields comparable R^2 values, confirming that the null result is not a statistical artifact of sparse coverage.

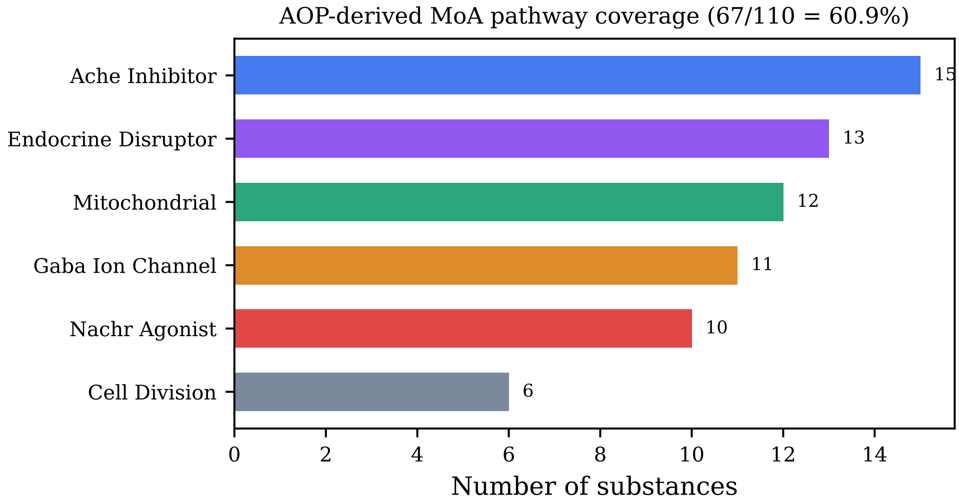

Figure 7: AOP-Wiki-derived mechanism-of-action pathway coverage across the 110 pesticide dataset. 67 substances (60.9%) map to at least one of 6 pathway categories.

4 Discussion

4.1 Why Does QML Fail on DegT50?

The negative R^2 of the VQC for DegT50 prediction is not a failure of quantum computing per se, but rather a predictable consequence of applying a 153–301-parameter model to 110 data points. Several factors contribute:

1. **Overfitting:** Despite the improved $N_{\text{params}}/N_{\text{data}}$ ratio of 1.39 for the 8-qubit DegT50 circuit (vs. 3.42 for the original 12-qubit circuit), the VQC still produces negative R^2. Even near-unity ratios appear insufficient for generalization with $N = 110$ samples. Classical models with far fewer effective parameters (RF uses bagging and feature subsampling as implicit regularization) are more robust.

2. **Barren plateaus:** With 8–12 qubits and 6–8 layers, the VQC may suffer from vanishing gradients in portions of the parameter landscape,[21] making optimization unreliable with limited training data for validation.

3. **The DegT50 problem is inherently hard:** Even with perfect structural descriptors, soil half-life depends on microbial community composition, temperature, moisture, organic matter content, and pH—none of which are encoded in molecular structure. An R^2 ceiling near 0.2–0.3 may be fundamental for structure-only models.

4. **Linear readout bottleneck:** The VQC readout layer is a linear combination of $\langle Z_i \rangle$ expectation values. Any function the circuit can express must therefore lie in the span of these N_{qubits} observables. Given that Ridge regression—itself a linear model—already achieves $R^2 = 0.825$ for K_{oc} with the 21-feature set (inflated by bioaccessibility circularity; the fair 20-feature R^2 is 0.532), and that the fair VQC achieves only $R^2 = 0.231$, the IQP-style quantum feature map appears to *reduce* rather than enhance the linear decodability of the molecular descriptor features. The classical descriptors, when passed directly to Ridge regression, retain more predictive structure than after transformation through the quantum embedding.

To disentangle these hypotheses, we conducted a systematic architecture search testing four circuit variants via 5-fold CV on DegT50 only (table 3): (A) a shallow-wide circuit (12 qubits, 3 layers, all-to-all CZ entanglement, double data re-uploading); (B) a deep-narrow circuit (8 qubits, 12 layers, ring CNOT, triple re-uploading); (C) a hardware-efficient template using PennyLane's `StronglyEntanglingLayers`; and (D) the baseline architecture with a reduced learning rate (0.03) and extended training (120 epochs).

Table 3: Architecture experiment: DegT50 prediction (5-fold CV) across four circuit variants. These experiments used the initial single 12-qubit circuit ($R^2 = -0.141$); the per-target 8-qubit circuit ($R^2 = -0.028$) was developed subsequently based on these findings.

Variant	**Architecture**	N_q	L	**Epochs**	**Train MSE**	$\mathbf{R^2}$	**Time**
A	All-to-all CZ, 2× re-upload	12	3	80	0.183	−0.157	2.1 h
B	Ring CNOT, 3× re-upload	8	12	100	0.003	−0.190	3.3 h
C	`StronglyEntanglingLayers`	12	5	60	0.165	−0.146	1.3 h
D	Baseline+, lr=0.03	12	8	120	0.009	−0.205	4.1 h
Current baseline (lr=0.04)		12	8	60	—	−0.141	

No variant improves upon the baseline. Remarkably, the two variants that achieve near-zero training MSE—variant B (0.003) and variant D (0.009)—produce the *worst* test R^2 (−0.190 and −0.205, respectively), while the underfit variants A and C (training MSE $\sim$0.17) perform marginally better. This anti-correlation between training fitness and generalization is a textbook signature of severe overfitting and demonstrates that the VQC has *sufficient* expressivity to memorize DegT50 training data but cannot extract a generalizable structure–activity relationship. The failure is therefore intrinsic to the target property and feature set, not to circuit architecture. Notably, variant A uses only 3 layers ($N_{\text{params}}/N_{\text{data}} \approx 1.0$), yet still produces negative R^2, demonstrating that reducing the parameter count alone does not rescue DegT50 prediction.

4.2 When Might QML Add Value?

Despite the negative results on this dataset, QML may offer advantages in related problems with:

- Larger datasets ($N > 1000$) where the parameter-to-data ratio is favorable
- Higher-dimensional feature spaces where quantum feature maps may provide kernel advantages [9]
- Quantum chemical descriptors (orbital energies, electron correlations) that naturally map to qubit representations. For example, the N_D electron correlation descriptor derived from coupled-cluster natural orbital occupations [22] quantifies multireference character across molecular systems and could serve as a qubit-native feature encoding, replacing or augmenting classical RDKit descriptors with information rooted in the electronic wavefunction.

4.3 The Value of Negative Results

This study contributes to the growing body of critical QML benchmarks that distinguish genuine quantum advantage from overstatement. Reporting negative results is essential for the field's credibility and helps practitioners avoid deploying quantum solutions where classical methods suffice.[8]

5 Conclusions

We presented a comparative study of variational quantum circuits and classical machine learning for predicting pesticide environmental fate properties from the EU Pesticides Database database. Key findings:

1. **Per-target circuit design reduces overfitting:** Replacing the single 12-qubit circuit ($N_{\text{params}}/N_{\text{train}} = 3.42$) with an 8-qubit DegT50 circuit ($N_{\text{params}}/N = 1.74$) improved DegT50 R^2 from -0.141 to -0.028 (5-fold CV). However, the R^2 remains below zero, meaning the model still fails to outperform a constant mean prediction.

2. **Classical models dominate:** Random Forest achieves $R^2 = 0.287$ for DegT50 and 0.712 for K_{oc} (LOO-CV, fair 20-feature set excluding bioaccessibility). Gradient Boosting performs comparably (0.283, 0.750), while the MLP neural network (0.075, 0.497) underperforms, confirming that high-capacity models struggle with small datasets.

3. **Hybrid stacking provides no significant improvement:** Nested LOO-CV for leakage-free α optimization yields $\alpha = 0.20 \pm 0.02$, producing a hybrid DegT50 $R^2 = 0.288$—an improvement of just +0.001 over standalone RF. This margin is negligible. The VQC does not capture complementary patterns beyond what the 21-feature RF already encodes.

4. **DegT50 is structurally bounded:** All models achieve $R^2 \leq 0.29$ (LOO-CV), supporting the hypothesis that soil degradation depends on environmental factors (microbial community, temperature, moisture) not captured by molecular structure. An R^2 ceiling near 0.3 may be fundamental for structure-only models.

5. **Minimum dataset recommendation:** Based on the observed failure at $N = 110$ and the parameter-to-data ratio analysis, we estimate that VQC generalization for circuits of this size requires $N \gtrsim 1000$ samples—an order of magnitude beyond current regulatory databases.

These results caution against premature claims of quantum advantage on small regulatory datasets. Even with per-target circuit design, early stopping, and hybrid stacking with proper nested cross-validation, the VQC does not outperform a well-engineered Random Forest. Negative results such as this are essential for grounding the rapidly growing QML field in empirical reality.

Acknowledgments

All quantum circuit simulations were performed using the PennyLane statevector simulator on consumer-grade hardware. Classical baseline computations used scikit-learn. The author thanks the PennyLane and scikit-learn developer communities for their open-source contributions.

Data and Software Availability

The substance database, quantum circuit implementation, classical baselines, and all analysis code are available at:
Zenodo (DOI: `https://doi.org/10.5281/zenodo.19386784`) and are also available on GitHub:
`https://github.com/c-arda/Quantum-Pesticide-Fate-Modeling`

Conflicts of Interest

The author declares no competing interests.

References

[1] European Parliament and Council. Regulation (EC) No 1107/2009 of the European Parliament and of the Council concerning the placing of plant protection products on the market. *Official Journal of the European Union*, L 309:1–50, 2009. OJ L 309, 24.11.2009.

[2] FOCUS Work Group. FOCUS groundwater scenarios in the EU review of active substances. Technical report, European Commission, SANTE/10161/2005, 2006. Report of the FOCUS Work Group on Groundwater Scenarios, Version 2.

[3] OECD. OECD guideline for the testing of chemicals, test no. 307: Aerobic and anaerobic transformation in soil. Technical report, Organisation for Economic Co-operation and Development, Paris, 2002.

[4] OECD. OECD guideline for the testing of chemicals, test no. 106: Adsorption – desorption using a batch equilibrium method. Technical report, Organisation for Economic Co-operation and Development, Paris, 2000.

[5] Leo Breiman. Random forests. *Machine Learning*, 45:5–32, 2001. doi: 10.1023/A:1010933404324.

[6] Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232, 2001. doi: 10.1214/aos/1013203451.

[7] Kamel Mansouri, Chris M. Grulke, Richard S. Judson, and Antony J. Williams. Opera models for predicting physicochemical properties and environmental fate endpoints. *Journal of Cheminformatics*, 10(10):1–15, 2018. doi: 10.1186/s13321-018-0263-1.

[8] Marco Cerezo, Andrew Arrasmith, Ryan Babbush, Simon C. Benjamin, Suguru Endo, Keisuke Fujii, Jarrod R. McClean, Kosuke Mitarai, Xiao Yuan, Lukasz Cincio, and Patrick J. Coles. Variational quantum algorithms. *Nature Reviews Physics*, 3:625–644, 2021. doi: 10.1038/s42254-021-00348-9.

[9] Vojtěch Havlíček, Antonio D. Córcoles, Kristan Temme, Aram W. Harrow, Abhinav Kandala, Jerry M. Chow, and Jay M. Gambetta. Supervised learning with quantum-enhanced feature spaces. *Nature*, 567:209–212, 2019. doi: 10.1038/s41586-019-0980-2.

[10] Maria Schuld, Alex Bocharov, Krysta M. Svore, and Nathan Wiebe. Circuit-centric quantum classifiers. *Physical Review A*, 101:032308, 2020. doi: 10.1103/PhysRevA.101.032308.

[11] Adrián Pérez-Salinas, Alba Cervera-Lierta, Elies Gil-Fuster, and José I. Latorre. Data re-uploading for a universal quantum classifier. *Quantum*, 4:226, 2020. doi: 10.22331/q-2020-02-06-226.

[12] Maria Schuld, Ryan Sweke, and Johannes Jakob Meyer. Effect of data encoding on the expressive power of variational quantum machine-learning models. *Physical Review A*, 103:032430, 2021. doi: 10.1103/PhysRevA.103.032430.

[13] Ville Bergholm, Josh Izaac, Maria Schuld, Christian Gogolin, Shahnawaz Ahmed, Vishnu Ajber, M. Sohaib Alam, Guillermo Alonso-Linaje, B. AkashNarayanan, Ali Asber, et al. PennyLane: Automatic differentiation of hybrid quantum-classical computations. *Quantum*, 6:660, 2022. doi: 10.22331/q-2022-03-30-660.

[14] EU pesticides database. https://food.ec.europa.eu/plants/pesticides/eu-pesticides-database_en, 2024. Accessed: 2025.

[15] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *Proceedings of the 3rd International Conference on Learning Representations (ICLR)*, 2015. arXiv:1412.6980.

[16] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[17] Arthur E. Hoerl and Robert W. Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970. doi: 10.1080/00401706.1970.10488634.

[18] Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, 58(1):267–288, 1996. doi: 10.1111/j.2517-6161.1996.tb02080.x.

[19] AOP-Wiki: Adverse outcome pathway knowledge base. https://aopwiki.org/, 2024. Accessed: 2025.

[20] Daniel L. Villeneuve, Doug Crump, Natalia Garcia-Reyero, Markus Hecker, Thomas H. Hutchinson, Carlie A. LaLone, Brigitte Landesmann, Teresa Lettieri, Sharon Munn, Malgorzata Nepelska, et al. Adverse outcome pathway (AOP) development I: Strategies and principles. *Toxicological Sciences*, 142(2):312–320, 2014. doi: 10.1093/toxsci/kfu199.

[21] Jarrod R. McClean, Sergio Boixo, Vadim N. Smelyanskiy, Ryan Babbush, and Hartmut Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9:4812, 2018. doi: 10.1038/s41467-018-07090-4.

[22] Celal Arda. Natural orbital correlation analysis of cluster bonding: From aromatic clusters to metallic superatoms with quantum topology probes, 2026. Unpublished work.